Bowen Xie

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RESEARCH INSTERESTS

Applied Probability, Queueing Theory, Stochastic Control, Operations Research, Mathematical Finance, Reinforcement Learning, Numerical Analysis

EDUCATION

Doctor of Phiolosophy — **Iowa State University**, Department of Mathematics — Aug 2017 -Aug 2022
- Applied Mathematics, GPA: 3.92/4.00
- Advisor: Dr. Ananda Weerasinghe

Master of Science — **Iowa State University**, Department of Mathematics — Aug 2017 -May 2019
- Applied Mathematics, GPA: 3.90/4.00
- Advisor: Dr. Ananda Weerasinghe

Bachelor of Science — **Liaoning Technical University**, Department of Mathematics — Sep 2013 -July 2017
- Mathematics & Applied Mathematics, GPA: 3.51/4.00 (Math GPA: 3.77/4.00)

EXPERIENCE

The University of Akron, DATA+ Faculty Fellow — Akron, OH
Institute for Data, Analytics, and Technology Advancement — Fall 2026-present

The University of Akron, Assistant Professor — Akron, OH
Department of Mathematics, College of Engineering & Polymer Science — Fall 2024-present

Washington University in St. Louis, Postdoctoral Lecturer — St. Louis, MO
Department of Statistics and Data Science — Fall 2022-Spring 2024

Iowa State University, Graduate Teaching/Research Assistant — Ames, IA
Department of Mathematics — Fall 2017-Summer 2022

MENTORING AND STUDENTS

Niraj Jaishwal — Akron, OH
- Undergraduate student from Computer Science — Fall 2025-present

Samuel Adjei Baiden — Akron, OH
- Master student from Applied Mathematics — Fall 2024-Spring 2026

Benno Lutz — Akron, OH
- Undergraduate student honors project — Fall 2025-Spring 2026

PUBLICATIONS AND PREPRINTS

Publications

[15] Y. Bai, Y. Gao, B. Xie, and S. Luo, "The symmetry method for fractional airy equation," *Mathematical Methods in the Applied Sciences*, vol. 49, no. 14, pp. 16 072–16 087, 2026.

[14] B. Xie and Y. Gao, "Numerical analysis for fractional riccati differential equations based on finite difference method," *Journal of Nonlinear Modeling and Analysis https://journal. global-sci. org/jnma*, vol. 7, no. 1, pp. 189–208, 2025.

[13] B. Xie, "Multi-component matching queues in heavy traffic," *Queueing Systems*, pp. 1–47, 2024, (Part of a Special Issue: Product Forms, Stochastic Matching, and Redundancy).

[12] B. Xie and Y. Gao, "Diffusion approximation of a special bandwidth sharing model via infinitesimal generators with a lifting-projection method," *Operations Research Letters*, vol. 52, p. 107 063, 2024.

[11] B. Xie and Y. Gao, "On the long-run average cost minimization problem of the stochastic production-inventory model," *Journal of Industrial and Management Optimization*, vol. 20(5), pp. 1823–1844, 2024.

[10] B. Xie and Y. Gao, "Population growth model with fractional operator and its numerical analysis," *Journal of Mathematical Study*, 2024, (to be appeared).

[9] Y. Gao and B. Xie, "Numerical analysis for fractional bratu type equation with explicit and implicit methods," *Mathematical Methods in the Applied Sciences*, vol. 46, no. 17, pp. 18 447–18 457, 2023, (corresponding author).

Preprints and Working Papers

[8] S. Adjei Baiden, B. Xie, and Y. Gao, *Optimal dynamic pricing under double-ended reflected stochastic market potential*, (under review at IISE Transactions).

[7] B. Xie, J. Chávez-Casillas, and J. E. Figueroa-López, *Market making in limit order books*, (Working paper).

[6] S. Ray, R. Wu, and B. Xie, *Regulated multi-component matching queues with abandonment: Scaling limits and large deviations*, (near completion), 2026.

[5] B. Xie and Y. Gao, *Joint pricing and innovation control in regulated recycling-rate diffusion*, (submitted to IISE Transactions), 2026.

[4] B. Xie, *An infinitesimal generator approach on weak convergence of regulated multi-class matching systems*, arXiv preprint arXiv:2507.09789 (submitted to Journal of Applied Probability, R&R), 2025.

[3] B. Xie and H. Yin, *Admission control for a single server waiting time process in heavy traffic*, arXiv preprint arXiv:2212.05674 (to be submitted to Queueing Systems), 2025.

Theses

[2] B. Xie, "Topics of queueing theory in heavy traffic (ph. d thesis)," `https://dr.lib.iastate.edu/items/be350d6c-e38a-4793-8e40-33445c2cdb5f/full`, Ph.D. dissertation, 2022.

[1] B. Xie, "Long-run average cost minimization of a stochastic processing system (master creative component)," `https://dr.lib.iastate.edu/entities/publication/3f5cb727-8a92-46c2-8343-9252e1e7c78c`, M.S. thesis, 2019.

SELECTED RESEARCH PROJECTS

Dynamics of limit order books in high-frequency trading [7] Fall 2023-present

Postdoc Mentor: Dr. José E. Figueroa-López, Washington University in St. Louis
Joint work with Dr. Jonathan Chávez Website, University of Rhode Island

- Construct a matching-system framework for Limit Order Books (LOBs) and develop algorithms for the optimal placement problem faced by market makers.
- Study the performance of reinforcement learning (RL) in solving market-making problems with inventory risk.
- Attempt to train market-making agent using RL with a real-world stock mar-

ket constructed from historical Limit Order Book (LOB) data.

- Attempt to conduct Neural Networks to approximate state-value and state-action value functions and compare their performance with tile coding as well as the theoretical solution to the optimal placement problem.

Joint stochastic control on product and process innovations [5] Spring 2025-present

- Production and process innovation are key drivers of quality improvement and cost reduction. In the circular economy, dynamic demand and resource recycling are crucial for operational efficiency.
- We explore stochastic control strategies within the propagation of chaos framework. The joint control of the product price and innovations under constraints such as recycling rates and demand propagation will be addressed.
- Our approach integrates rigorous stochastic analysis to develop adaptive and tractable solutions for complex systems, which are applicable to Monopoly's product and process innovations.

Scaling-limits on regulated matching systems [6] Fall 2024-present
Joint work with Dr. Souvik Ray and Dr. Ruoyu Wu, Iowa State University

- Construct a diffusion approximation for a large-scale regulated matching system, characterized by a coupled regulated stochastic integral equation.
- Provided a non-trivial approach to show the existence and uniqueness of the weak solution to such non-Lipschitz coupled regulated stochastic integral equations and simulated the matching behavior.
- Developing control strategies for heterogeneous matching systems, especially under resource constraints requiring admission or buffer policies.

Multi-component Matching Queue Systems in Heavy Traffic [13] Summer 2019-2022
Advisor: Dr. Ananda Weerasinghe, Iowa State University

- Constructed many-component matching queue models and studied the diffusion-scaled heavy traffic limits, which are characterized by a coupled non-linear stochastic integral equation.
- Established the convergence of the cost functional under cost structures.
- Formulated an asymptotic relationship between the queue length processes and the virtual waiting time processes (the asymptotic Little's law) and further refined the result to L^p sense on a special probability space.
- Simulated the matching queue systems and the corresponding stochastic integral equations numerically in Python.
- Observe that a matching queue model without abandonment could possibly be interpreted as a basket option pricing problem.

INVITED TALKS

INFORMS Annual Meeting 2026 San Francisco, CA
Scaling limits for regulated multi-component matching queues with abandonment November 1-4, 2026

Stochastic Network Conference 2026, Poster presentation Chicago Booth, IL
Scaling limits of regulated multi-component matching systems June 14-19, 2026

SIAM Math Club Colloquium, at The University of Akron Akron, OH
Inside High-Frequency Trading: The Mathematics of the Limit Order Book Mar 13, 2026

INFORMS Annual Meeting 2025 Atlanta, GA
From a single-server waiting time control to reinforcement learning motivation for matching systems Oct 26-29, 2025

INFORMS Applied Probability Society Conference 2025 Atlanta, GA
Admission control for a single server waiting time process in heavy traffic, June 30-July 3, 2025

Intro to regulated matching systems

SIAM Math Club Colloquium, at The University of Akron — Akron, OH
Stochastic Control in Action: Theory, Applications, and Reinforcement Learning — Mar 21, 2025

Colloquium for Pure and Applied Math, at The University of Akron — Akron, OH
Queueing Theory and Stochastic Controls:
Applications in Industry, Economics, and Financial Math — Oct 11, 2024

2024 Joint Mathematics Meetings, AMS Special Session — San Francisco, CA
Dynamics of Matching Queues in Heavy Traffic and
Some New Findings of Regulated Matching Systems — Jan 3-6, 2024

2023 INFORMS Annual Meeting, Applied Probability Society Cluster — Phoenix, AZ
Multi-component matching and its dynamics and some new findings — Oct 15-18, 2023

Statistics and Data Science Seminar, at Washington University in St. Louis — St. Louis, MO
Many-component Matching Queue Systems in Heavy Traffic — Sep 21, 2022

SIAM Graduate Student Seminar, at Iowa State University — Ames, IA
Multi-component Matching Queues in Heavy Traffic — Apr 08, 2022

Probability and Statistics Seminar, at University of Wisconsin-Milwaukee — Virtual
Many-component Matching Queue System — Dec 1, 2021

20th Northeast Probability Seminar — Virtual
Many-component Matching Queue System — Nov 18, 2021

SIAM Graduate Student Seminar, at Iowa State University — Virtual
Optimal Control Problem for a Diffusion Process — Nov 06, 2020

COMMITTEES AND PROFESSIONAL SERVICE

INFORMS Education Outreach Graduate Committee — Akron, OH
Organize webinars and in-person panel discussions at the INFORMS Annual Meeting to promote graduate education. — Fall 2025-present

Reviewer, AMS Mathematical Reviews/MathSciNet Reviewer — Akron, OH
Reviewed scholarly papers for inclusion in the AMS Mathematical Reviews database and Mathematics Subject Classification. — Fall 2024-present

Reviewer, Methodology and Computing in Applied Probability — Akron, OH
Provide peer reviews for submitted manuscripts. — Fall 2024-present

Reviewer, SIAM Journal on Financial Mathematics — Akron, OH
Provide peer reviews for submitted manuscripts. — Fall 2024-present

Graduate Admission Committee, at The University of Akron — Akron, OH
Review graduate student applications and assistantship applications in Math Department. — Fall 2024-present

Administrative Assistant — Ames, IA
Assisted the Math Department with administrative tasks, including library organization, website updates, and the Math Genealogy Project. — Summer 2021

SIAM Student Chapter — Ames, IA
Served as treasurer and participated in the planning committee for the SIAM Graduate Student Seminar. — Fall 2020-Spring 2021

CONFERENCE ATTENDANCE

INFORMS Annual Meeting 2026, *San Francisco, CA* — November 1-4, 2026

Stochastic Network Conference 2026, *Chicago Booth, IL* — June 15-19, 2026

INFORMS Annual Meeting 2025, *Atlanta, GA*	Oct 26-29, 2025
SNAPP Summer School on Stein's Method for Queueing Approximations, *Virtual*	July 29-Aug 28, 2025
INFORMS APS Summer School, *Atlanta, GA*	June 25-29, 2025
INFORMS Applied Probability Society Conference 2025, *Atlanta, GA*	June 30-July 3, 2025
Joint Mathematics Meetings 2024, *San Francisco, CA*	Jan 2024
INFORMS Annual Meeting 2023, *Phoenix, AZ*	Oct 15-18, 2023
Seminar on Stochastic Processes, *Tucson, AZ*	Mar 8-12, 2023
First-Passage Percolation and Related Models Program, *ICTS, India (Hybrid)*	July 11-29, 2022
6th Annual Meeting of SIAM Central States Section, *Virtual*	Oct 2021
5th Annual Meeting of SIAM Central States Section, *Ames, IA*	Oct 2019
41st Midwest Probability Colloquium, *Evanston, IL*	Oct 2019

TEACHING EXPERIENCE

Teaching Experience at The University of Akron

Asst Professor, MATH 149 - Precalculus, *11 students*	Fall 2026
Asst Professor, MATH 335 - Intro to Ordinary Diff Equ, *44 students*	Fall 2026
Asst Professor, MATH 300 - Tools for Data Science, *16 students*	Spring 2026
Asst Professor, MATH 450/550 - Optimization, *8 undergrads, 5 grads*	Spring 2026
Asst Professor, MATH 335 - Intro to Ordinary Diff Equ, *42 students*	Fall 2025
Asst Professor, MATH 221 - Analytic Geometry-Calculus I, *34 students*	Fall 2025
Asst Professor, MATH 450/550 - Optimization, *9 students*	Spring 2025
Asst Professor, MATH 221 - Analytic Geometry-Calculus I, *40 students*	Spring 2025
Asst Professor, MATH 335 - Intro to Ordinary Diff Equ, *40 students*	Fall 2024
Asst Professor, MATH 221(Honors) - Analytic Geometry-Calculus I, *26 students*	Fall 2024

Teaching Experience at Washington University in St. Louis

Postdoctoral Lecturer, MATH 495/5070 - Stochastic Processes, *27 students*	Spring 2024
Postdoctoral Lecturer, MATH 493/5010 - Probability, *40 students*	Spring 2024
Postdoctoral Lecturer, MATH 5010 - Probability, *50 students*	Fall 2023
Postdoctoral Lecturer, MATH 493 - Probability, *8 students*	Summer 2023
Postdoctoral Lecturer, MATH 493 - Probability, *73 students*	Spring 2023
Postdoctoral Lecturer, MATH 3200 - Statistics and Data Analysis, *70 students*	Fall 2022
Postdoctoral Lecturer, MATH 493 - Probability, *48 students*	Fall 2022

Teaching Experience at Iowa State University

Graduate Teaching Assistant, MATH 645 - Advanced Stochastic Processes	Spring 2022
Graduate Teaching Assistant, MATH 481/581 - Numerical Methods for Diff Equ	Spring 2022
Graduate Teaching Assistant, MATH 517 - Finite Difference Methods	Spring 2022
Graduate Teaching Assistant, MATH 414 - Analysis I	Spring 2022
Graduate Teaching Assistant, MATH 554 - Intro to Stochastic Processes	Fall 2021
Graduate Teaching Assistant, MATH 561 - Numerical Analysis	Fall 2021
Recitation Instructor, MATH 143 - Preparation for Calculus	Spring 2021
Recitation Instructor, MATH 143 - Preparation for Calculus	Fall 2020
Recitation Instructor, MATH 150 - Discrete Mathematics	Spring 2020
Recitation Instructor, MATH 143 - Preparation for Calculus	Fall 2019
Lecturer, MATH 150 - Discrete Mathematics for Business and Social Sciences	Summer 2019
Recitation Instructor, MATH 151 - Calculus for Business and Social Sciences	Spring 2019
Graduate Teaching Assistant, MATH 554 - Introduction to Stochastic Processes	Fall 2018

Graduate Teaching Assistant, MATH 435 - Geometry I	Fall 2018
Graduate Teaching Assistant, MATH 373 - Introduction to Scientific Computing	Fall 2018
Graduate Teaching Assistant, MATH 104 - Introduction to Probability	Spring 2018
Graduate Teaching Assistant, MATH 304 - Combinatorics	Fall 2017
Graduate Teaching Assistant, MATH 201 - Introduction to Proofs	Fall 2017

GRADUATE COURSES

- Methods of Applied Math I, II
- Ordinary Differential Equation and Dynamic Systems
- Introduction to Stochastic Processes
- Advanced Stochastic Processes
- Machine Learning
- Numerical Analysis
- Numerical Linear Algebra
- Finite Difference Methods
- Advanced Special Topics: Continuous Time Asset Pricing Models
- Optimal Transport and Deep Learning
- Theory and Methods for Continuous Optimization (Audit)

OTHER PROJECTS

These projects are done for the laboratory parts of the Machine Learning course. Spring 2020

- **Naive Bayes Classifier for Text Classification.** Implement the Naive Bayes algorithm to tackle the "20 Newsgroups" classification problem.
- **Decision Tree Classifier.** Experiment with the Decision Tree classifier on the congressional voting records data set using Weka.
- **Neural Network Classifier.** Using the TensorFlow package to perform an experiment on the optical recognition of the handwritten digits data set.
- **Ensemble Learning.** Train ensemble models, random forest, and AdaBoost.M1, and construct an ensemble classifier by training 4 individual models on the blood transfusion service center data set.